

CS 513: Discrete Mathematics

Course Code	CS 513
Course Name	Discrete Mathematics
L-T-P-C	3-1-0-4
Prerequisites	none
Intended for	MTech CSE
Mutual Exclusion	CS208 and CS511.
Approval	44th BoA (Nov 2021)

Course Contents

- **Logic:** Propositional logic syntax and semantics (revision); proof system and deduction; soundness and completeness; principle of resolution; (ordered) binary decision diagrams; first order logic syntax and semantics; structures, models, satisfaction and validity; resolution; unification; proof systems; axiomatization, soundness, completeness and incompleteness theorems; undecidability of validity problem. [8 Lectures]
- **Infinite and Structured Sets:** [12 Lectures]
 - Countable and uncountable sets, Cantor's diagonalization. Turing machines, Church- Turing Thesis. undecidability of the halting problem. consequences to the program verification problem.
 - **Abstract Algebra:** Homomorphism, Fundamental Theorem of homomorphisms, posets and lattices, formal contexts, monoids, semigroups, groupoids and groups, subgroups, cosets, Lagrange's theorem, rings, fields.
- **Combinatorics & Graph Theory:** [12 Lectures]
 - Counting arguments, recurrence relations, generating functions (Mnager problem), formal power series (ring).
 - Basics of graph theory (revision), planar graphs (Kuratowski's theorem), minor graphs (Wagner's theorem, Robertson-Seymour theorem) matching and covering (Hall's theorem, Tutte's theorem, Gallai/Milgram's theorem, connectivity and network flows, coloring (Brooks' theorem, Vizing's theorem), intersection and perfect graphs, sparse and dense graphs.
- **Probability Theory:** Recap of basic probability theory: axiomatic definition, discrete and continuous random variables, functions of random variables; joint, marginal, conditional distributions, Expectation and variance, Moment generating function and characteristic functions. Moments and deviations (Stable marriage problem, the coupon collectors problem), Concentration inequalities (Chernoff and Hoeffding bounds), Markov Chains and random walks (Expanders). Monte Carlo method. [10 Lectures]

Laboratory Modules

Tutorials on each of the above course modules (14 hours)

Text Books

1. K. H. Rosen, **Discrete Mathematics and Its Applications**, 8th Edition, McGraw Hill, 2019.
2. D. S. Dummit and R. M. Foote, **Abstract Algebra**, 3rd Edition, Wiley, 2004.
3. M. Mitzenmacher and E. Upfal, **Probability and Computing**, 2nd Edition, Cambridge Univ. Press, 2017.

References

- Dirk van Daalen, **Logic & Structure**, Springer, 2008.
- Uwe Schoening, **Logic for Computer Scientists**, Springer, 2008
- Michael Huth, Mark Ryan, **Logic in Computer Science: Modelling and Reasoning about Systems**, Cambridge University Press 2004.
- B. Ganter, R. Wille, **Formal Concept Analysis**, Springer, 1996.
- P. M. Cohn, **Universal Algebra**, Springer, 1981.
- W. Wechler, **Universal Algebra for Computer Scientists**, Springer, 1992.
- Reinhard Diestel, **Graph Theory**, 5th Edition, Springer, 2017.
- Bela Bollobas, **Modern Graph Theory**, Springer, 1998.